EXPERIMENT-05

ELECTROMAGNETIC SIMULATION AND S-PARAMETER CHARACTERIZATION OF A MICROWAVE DIRECTIONAL COUPLER AND TWO-WAY POWER DIVIDER

PROGRAM

```
clc;
clear;
close all;
% Microwave Semiconductor Devices and Mixer Analysis
figure('Color','w','Position',[100 50 1200 650]);
%% 1. Gunn Diode I-V Characteristics
subplot(3,3,1);
V = [0 3 6 10];
I = [0 15 9 15];
plot(V,I,'b-o','LineWidth',2,'MarkerSize',4);
grid on;
xlabel('Voltage (V)');
ylabel('Current (mA)');
title('Gunn Diode I-V Characteristics');
xlim([0 10]);
ylim([0 17]);
%% 2. Schottky Diode I-V Characteristics
subplot(3,3,2);
Vd = linspace(-0.3,0.6,500);
Is = 1e-12;
n = 1.5;
Vt = 0.02585;
Id = Is*(exp(Vd/(n*Vt))-1);
semilogy(Vd,max(abs(Id),1e-12),'r','LineWidth',2);
grid on;
```

```
xlabel('Voltage (V)');
ylabel('|Current| (A)');
title('Schottky Diode I-V Characteristics');
xlim([-0.3 0.6]);
ylim([1e-12 1e0]);
%% 3. PIN Diode Frequency Response
subplot(3,3,3);
f = linspace(0,20,300);
fc = 5;
response = -10*log10(1+(f/fc).^2);
plot(f,response,'m','LineWidth',2);
hold on;
xline(fc,'k--','Cutoff frequency');
grid on;
xlabel('Frequency (GHz)');
ylabel('Response (dB)');
title('PIN Diode Frequency Response');
xlim([0 20]);
ylim([-15 1]);
%% 4. Mixer IF Output Spectrum
subplot(3,3,4);
fIF = linspace(0,1.5,1000);
power = -155*ones(size(fIF));
fIF0 = 0.5;
power(abs(fIF-fIF0)<0.002) = -18;
plot(fIF,power,'b','LineWidth',1.5);
hold on;
plot(fIF0,-18,'ro','MarkerFaceColor','r');
xline(fIF0,'k--','Theoretical IF');
```

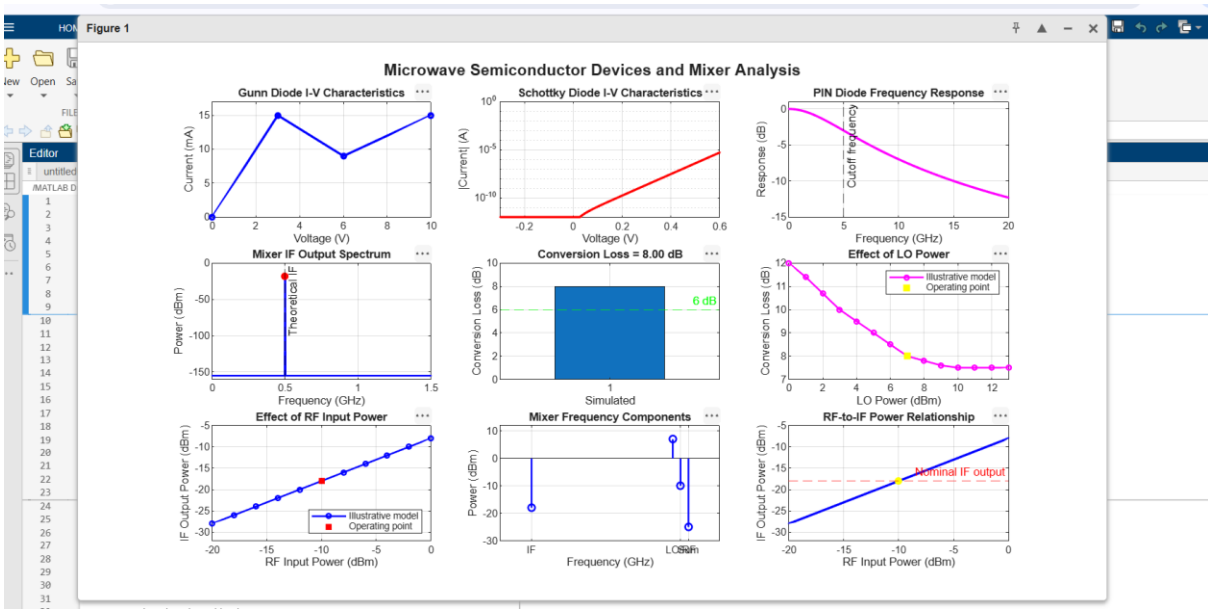
```
grid on;
xlabel('Frequency (GHz)');
ylabel('Power (dBm)');
title('Mixer IF Output Spectrum');
xlim([0 1.5]);
ylim([-160 0]);
%% 5. Mixer Conversion Loss
subplot(3,3,5);
conversionLoss = 8;
bar(1,conversionLoss,0.6);
hold on;
yline(6,'g--','6 dB');
grid on;
xlabel('Simulated');
ylabel('Conversion Loss (dB)');
title('Conversion Loss = 8.00 dB');
xlim([0.4 1.6]);
ylim([0 10]);
%% 6. Effect of LO Power
subplot(3,3,6);
LO = 0:1:13;
lossLO = [12 11.4 10.7 10.0 9.5 9.0 8.5 ...
          8.0 7.8 7.6 7.5 7.5 7.5 7.5];
plot(LO,lossLO,'mo-','LineWidth',1.5,...
    'MarkerSize',4);
hold on;
plot(7,8,'ys','MarkerFaceColor','y',...
    'MarkerSize',7);
grid on;
```

```
xlabel('LO Power (dBm)');
ylabel('Conversion Loss (dB)');
title('Effect of LO Power');
legend('Illustrative model','Operating point',...
    'Location','northeast');
xlim([0 13]);
ylim([7 12]);
%% 7. Effect of RF Input Power
subplot(3,3,7);
RF = -20:2:0;
IFout = RF - 8;
plot(RF,IFout,'bo-','LineWidth',1.5,...
    'MarkerSize',4);
hold on;
plot(-10,-18,'rs','MarkerFaceColor','r');
grid on;
xlabel('RF Input Power (dBm)');
ylabel('IF Output Power (dBm)');
title('Effect of RF Input Power');
legend('Illustrative model','Operating point',...
    'Location','southeast');
xlim([-20 0]);
ylim([-32 -5]);
%% 8. Mixer Frequency Components
subplot(3,3,8);
freqGHz = [0.5 9.5 10 10.5];
componentPower = [-18 7 -10 -25];
stem(freqGHz,componentPower,'b','LineWidth',1.5,...
    'Marker','o');
```

```
grid on;
xlabel('Frequency (GHz)');
ylabel('Power (dBm)');
title('Mixer Frequency Components');
xticks([0.5 10 10.5]);
xticklabels({'IF','LO/RF','Sum'});
ylim([-30 12]);
%% 9. RF-to-IF Power Relationship
subplot(3,3,9);
RF2 = -20:1:0;
IF2 = RF2 - 8;
plot(RF2,IF2,'b','LineWidth',2);
hold on;
yline(-18,'r--','Nominal IF output');
plot(-10,-18,'yo','MarkerFaceColor','y',...
    'MarkerSize',6);
grid on;
xlabel('RF Input Power (dBm)');
ylabel('IF Output Power (dBm)');
title('RF-to-IF Power Relationship');
xlim([-20 0]);
ylim([-32 -5]);
%% Overall Figure Title
sgtitle('Microwave Semiconductor Devices and Mixer Analysis',...
    'FontSize',14,'FontWeight','bold');
% Display results
fprintf('MICROWAVE SEMICONDUCTOR DEVICES AND MIXER ANALYSIS\n');
fprintf('-----------------------------------------------\n');
fprintf('PIN diode cutoff frequency = %.2f GHz\n',fc);
```

```
fprintf('Mixer IF frequency = %.2f GHz\n',fIF0);
fprintf('Mixer conversion loss = %.2f dB\n',conversionLoss);
fprintf('Selected LO power = 7 dBm\n');
fprintf('Selected RF input power = -10 dBm\n');
fprintf('Corresponding IF output power = -18 dBm\n');
fprintf('Simulation completed successfully.\n');
```

OUTPUT



RESULT

The simulation plots show the frequency-dependent behavior of the microwave directional coupler and two-way power divider over 9–11 GHz. These are illustrative behavioral models, not full electromagnetic simulations.